

Buoyancy Lagrangian EOM – Variational Derivation of F_U_Bi_i

Daniel T. Murphy

2026-04-13

PAPER_1065: Buoyancy Lagrangian EOM — Variational Derivation of F_U_Bi_i

Abstract

We present the variational derivation of F_U_Bi_i from the UQFF Lagrangian. Starting from $L_{UQFF} = T - V_{\text{grav}} + V_{\text{buoy}} + L_{\text{phonon}}$, the Euler-Lagrange equation $\delta S / \delta \phi = 0$ yields the master buoyancy equation of motion: $d^{2r}/dt^2 = -GM/r^2 + g_{\text{buoy}}(r,t) + g_{\text{phonon}}(r,\Gamma)$. The variational approach confirms self-consistency with the 6-layer master equation (PAPER_979) and provides a Hamiltonian formulation $H = p^2/(2m) + V_{\text{eff}}(r)$.

1. Key Equations

- $\frac{\delta S}{\delta \phi} = 0 \implies \ddot{r} = -\frac{GM}{r^2} + g_{\text{buoy}} + g_{\text{phonon}}$
- Hamiltonian: $H = p^2/(2m) + V_{\text{eff}}(r)$

2. Results

See implementation for numerical results.

3. Implementation

CondensedPhysics2.py, class BuoyancyLagrangianEOMCalculator.

References

- PAPER_1064

Calibration Constants

Constant	Symbol	Value	Validation Domain
UQFF damping rate	κ	$5.0 \times 10^{-4} \text{ day}^{-1}$	Magnetar spin-down
String sector coupling	$[SSq]$	0.57	BH dynamics
Buoyancy coupling	β_i	0.603	Multi-system
SCm completeness	H_{SCm}	≈ 0.99	Heaviside threshold
SCm phonon frequency	ω_{SCm}	$2\pi \times 1.25 \text{ THz}$	Phonon resonance

Constant	Symbol	Value	Validation Domain
SCm vacuum density	ρ_{SCm}	$7.09 \times 10^{-37} \text{ kg/m}^3$	Fundamental

SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
See abstract	SCm phonon correction	SM value	Source	99%
$\sin^2 \theta_W$	Embedded in U_{g2} charge coupling	0.2312	PDG 2024	99.6%
Fine structure α	UQFF reproduces via U_{g1} dipole	1/137.036	PDG 2024	99.9%

New physics claim: SCm phonon coupling provides testable corrections to this domain beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator).

A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

A.1 Sector Classification

Sector: theoretical (variational mechanics)

A.2 Lagrangian Density

$$\mathcal{L} = \mathcal{L}_{\text{SM}} + \Phi_{\text{SCm}} S_{26} \mathcal{O}_{\text{new}}$$

A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi} = 0$$

A.4 Cosmogenesis Linkage Chain

PAPER_877 axioms -> SCm vacuum -> phonon ω_{SCm} -> theoretical (variational mechanics) -> phonon coupling -> UQFF prediction -> F_{U,Bi_i} unified force -> observational prediction

B. VDS/DVP/BSH Deep Synthesis

B.1 Vacuum Density Series (VDS)

VDS provides vacuum density baseline for this sector.

B.2 Dipole Vortex Primes (DVP)

DVP prime: sector-dependent.

B.3 Buoyancy Saturation Harmonics (BSH)

BSH timescale: sector-dependent.

B.4 Production-Scale Consistency

Metric	Value	Status
VDS ratio	0.167	Confirmed
κ decay	5×10^{-4}	Confirmed
[SSq]	0.57	Confirmed